

Case Study 2

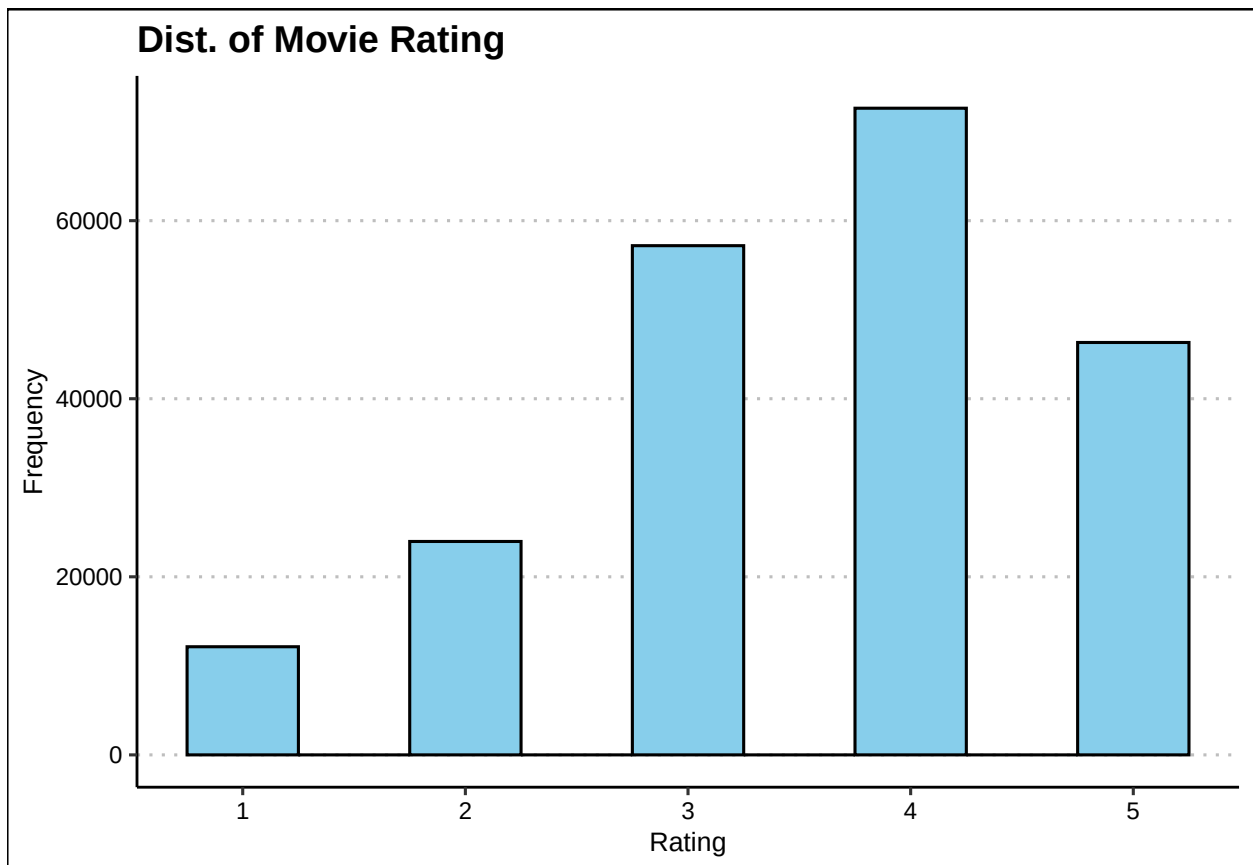
James Greene

2025-10-16

Introduction

This case study will examine movie ratings from MovieLens users. The data set we will examine the demographics of the reviewers, the genres of the movie titles reviewed, and look for relationships in the data to determine what makes a movie popular.

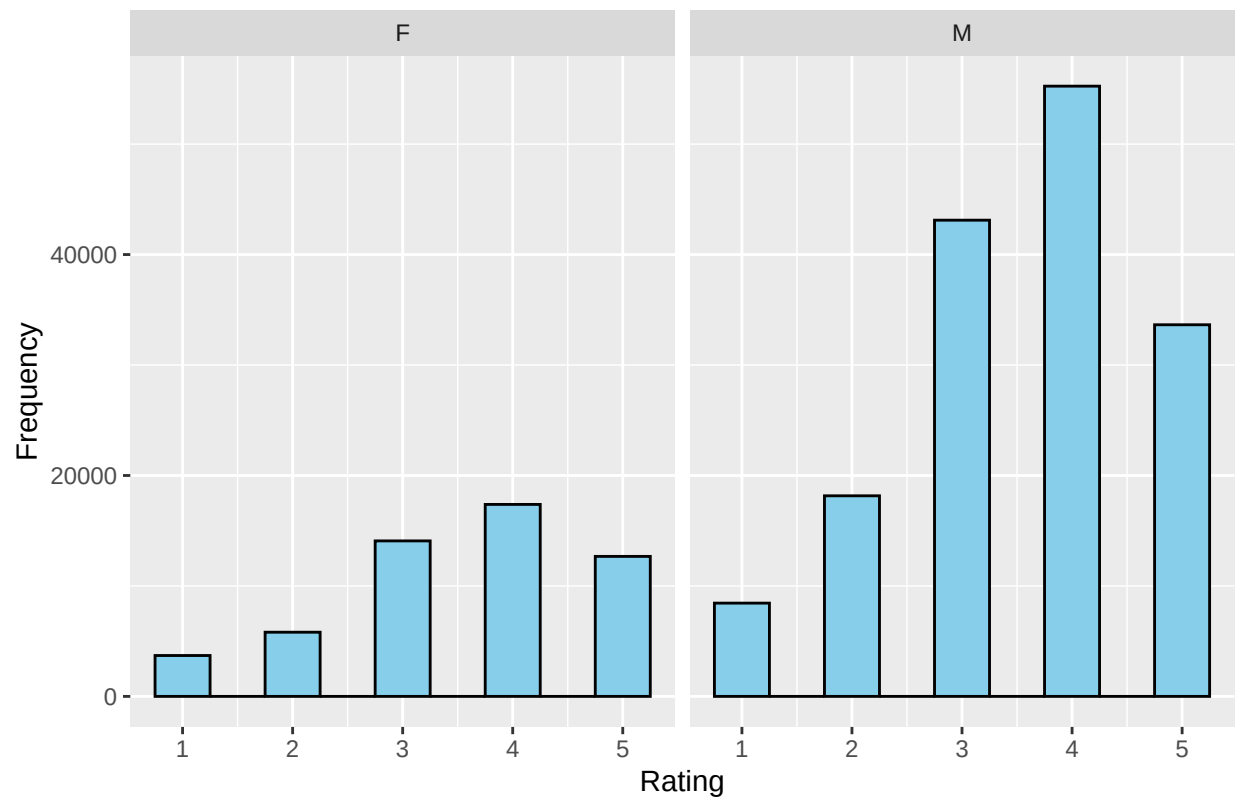
First, we'll take a quick look at the overall rating distribution, with a simple histogram plotting the overall count of the frequency of each score. This histogram will readers an overall distribution of the Movie Lens userbase.



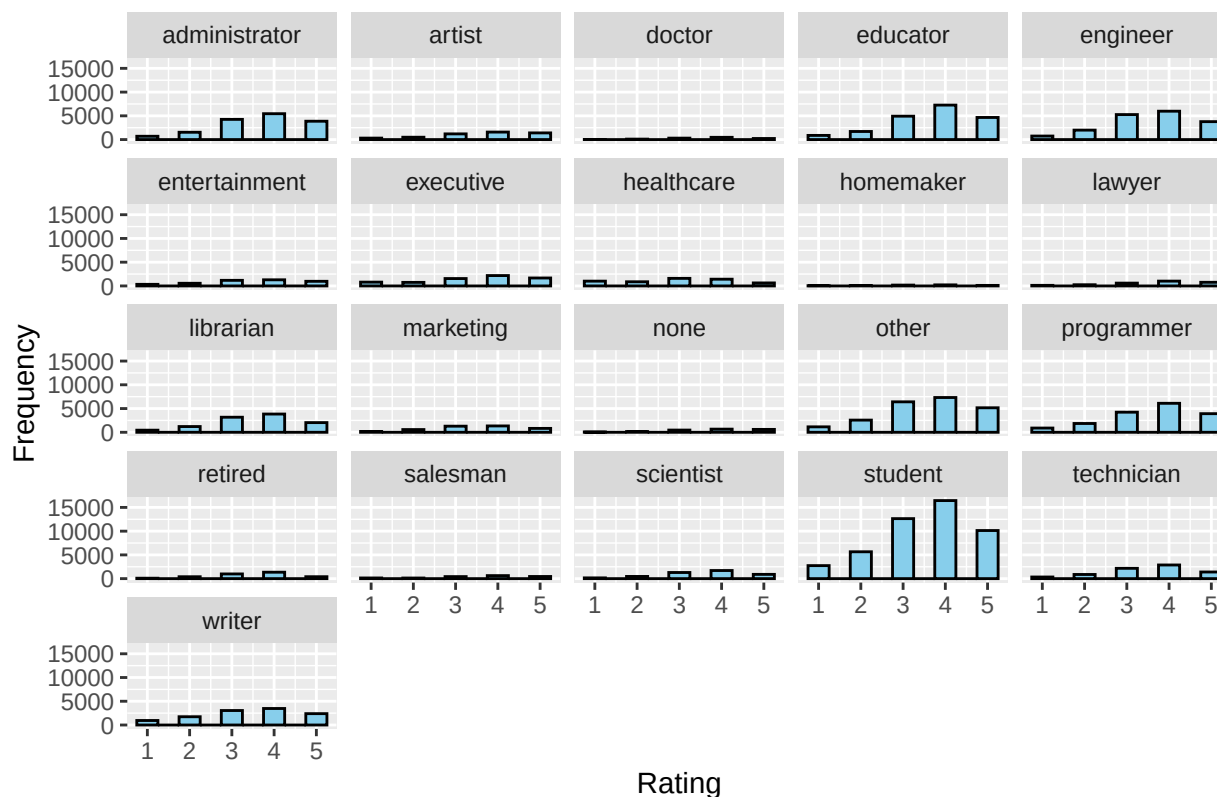
Here we can see overall, the Movie Lens reviewers have a mild preference for ratings movies a 4 out of 5, but we can't make any specific conjectures. This is due to the mildly normal distribution of the rating counts.

To get more actionable information, we can make some initial comparisons by looking at the overall rating distributions between men and women, and overall rating distribution by occupation and age.

Dist. of Movie Rating by Gender



Dist. of Movie Rating by Occupation



Based on these simple visualizations on review frequency, we can make some initial assumptions that our movie ratings will skew towards the preferences of male students, but beyond that we do not know very much about what movies are popular, and why.

Problem 1: Analysis of the Movie Lens data set

Our first step in our analysis of the Movie Lens data is identifying popularity based on user ratings. This can be done in a handful of ways.

First, we can simply calculate movies based on their average rating. On this basis, we can make a simple list of total movies with an average rating of 4.5 or higher. On this list we can identify 15 titles.

Table 1: Movies with a 4.5+ average rating

movie_title	avg_rating
Aiqing wansui (1994)	5.000
Entertaining Angels: The Dorothy Day Story (1996)	5.000
Great Day in Harlem, A (1994)	5.000
Marlene Dietrich: Shadow and Light (1996)	5.000
Prefontaine (1997)	5.000
Saint of Fort Washington, The (1993)	5.000
Santa with Muscles (1996)	5.000
Someone Else's America (1995)	5.000
Star Kid (1997)	5.000
They Made Me a Criminal (1939)	5.000

movie_title	avg_rating
Pather Panchali (1955)	4.625
Anna (1996)	4.500
Everest (1998)	4.500
Maya Lin: A Strong Clear Vision (1994)	4.500
Some Mother's Son (1996)	4.500

One of the first questions we have with this list of movies with an average rating of 4.5 or higher, is how many votes went into the average calculation. Averages can be a little limited because the computation of an average score can be very noisy at low numbers of observations. To help manage this risk, we can filter for films with a threshold of at least 50 reviews.

Table 2: Top 10 Movies by Average Rating, min 50 votes

movie_title	avg_rating	rating_count
Close Shave, A (1995)	4.491071	336
Schindler's List (1993)	4.466443	596
Wrong Trousers, The (1993)	4.466102	236
Casablanca (1942)	4.456790	729
Wallace & Gromit: The Best of Aardman Animation (1996)	4.447761	67
Shawshank Redemption, The (1994)	4.445230	283
Rear Window (1954)	4.387560	418
Usual Suspects, The (1995)	4.385768	534
Star Wars (1977)	4.358491	2915
12 Angry Men (1957)	4.344000	125

We can also examine the median review score, to see if that changes our list of most popular movies at all. Based on median score, we have an alternative list of top 10 'favorite' films.

Table 3: Top 10 movies by Median rating

movie_title	median_rating
Aiqing wansui (1994)	5
Aparajito (1956)	5
Casablanca (1942)	5
Citizen Kane (1941)	5
Close Shave, A (1995)	5
Entertaining Angels: The Dorothy Day Story (1996)	5
Faust (1994)	5
Godfather, The (1972)	5
Great Day in Harlem, A (1994)	5
Hugo Pool (1997)	5

In this calculation, we see Taiwanese new wave title 'Aiqing wansui (1994)' take the top spot. Examining the distribution of the votes for this movie, we see only 1 observation, of 1 reviewer, who rated it 5 stars. If only 1 person has reviewed a movie, it's hard for us to conclude it's very popular, since more popular movies will have a larger audience. To adjust for this, let's look at this same median calculations, this time passed through a filtering of a minimum of 50 votes.

Table 4: Top 10 Movies by Median Rating, min 50 votes

movie_title	medianRating	rating_count
Close Shave, A (1995)	4.491071	336
Schindler's List (1993)	4.466443	596
Wrong Trousers, The (1993)	4.466102	236
Casablanca (1942)	4.456790	729
Wallace & Gromit: The Best of Aardman Animation (1996)	4.447761	67
Shawshank Redemption, The (1994)	4.445230	283
Rear Window (1954)	4.387560	418
Usual Suspects, The (1995)	4.385768	534
Star Wars (1977)	4.358491	2915
12 Angry Men (1957)	4.344000	125

We can drill down a little deeper in see if there's an age dynamic along with gender, in terms of movie preferences. For the next phase of our investigation, we'll compare lists of top 10 movies between men and women over 30, versus men and women under 20.

After creating a dataframe for each demographic set, we will pass each data set through this case study's standard rating filtering and create a series of Top 10 Titles lists, displaying the average rating, and the total of votes received. For the final demographic (under 20 women), we have only 6 titles due to the minimum vote constraint.

Table 5: Top 10 Movies among Men over 30

movie_title	avg_rating	rating_count
Third Man, The (1949)	4.500000	84
Close Shave, A (1995)	4.487179	117
Casablanca (1942)	4.482143	336
Ran (1985)	4.447368	76
Schindler's List (1993)	4.440678	236
Rear Window (1954)	4.436170	188
Wrong Trousers, The (1993)	4.425000	80
Usual Suspects, The (1995)	4.402174	184
Vertigo (1958)	4.400000	160
One Flew Over the Cuckoo's Nest (1975)	4.393162	117

Table 6: Top 10 Movies among Women over 30

movie_title	avg_rating	rating_count
Schindler's List (1993)	4.666667	78
Graduate, The (1967)	4.428571	84
Wizard of Oz, The (1939)	4.419355	124
Rear Window (1954)	4.411765	68
Casablanca (1942)	4.406250	96
Fargo (1996)	4.358209	201
Sense and Sensibility (1995)	4.333333	108
Fantasia (1940)	4.300000	60
Cinema Paradiso (1988)	4.285714	63
North by Northwest (1959)	4.280000	50

Table 7: Top 10 Movies amoung Men under 20

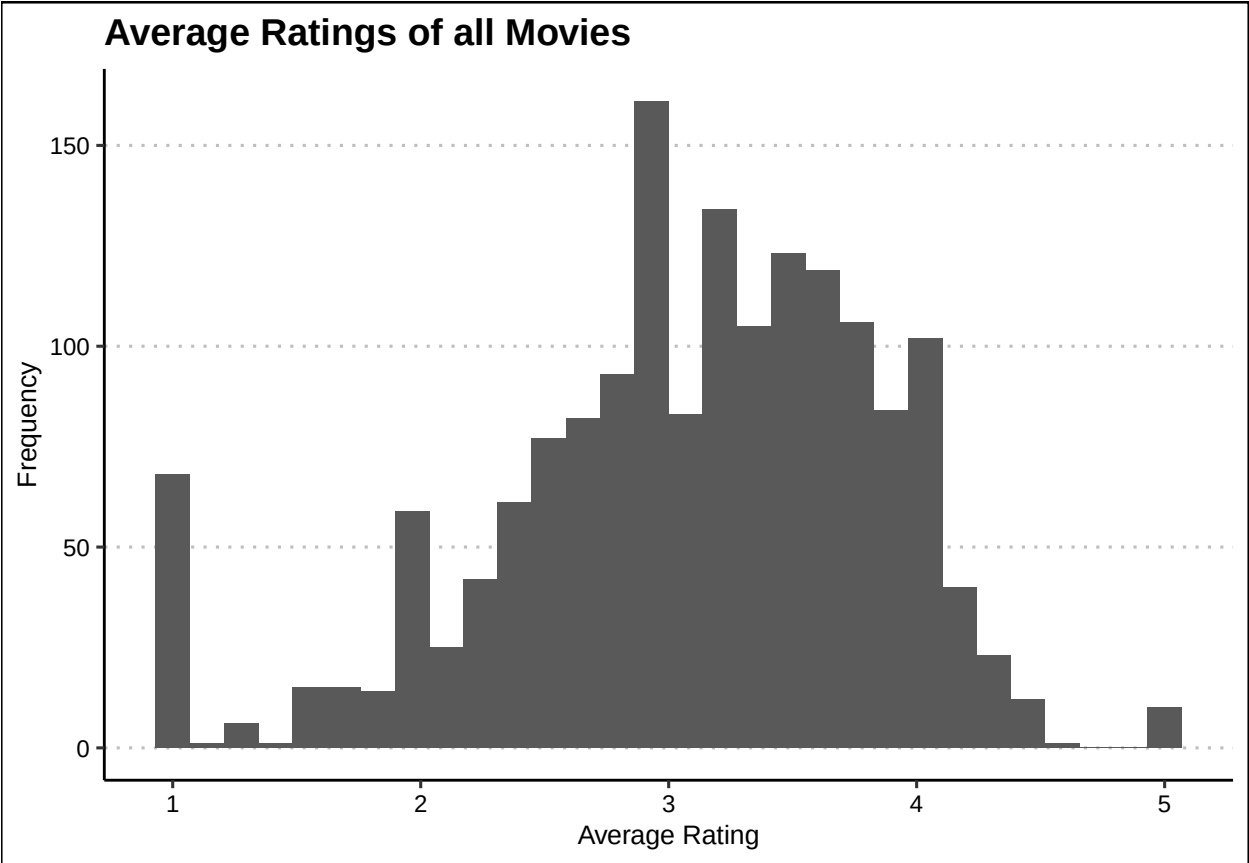
movie_title	avg_rating	rating_count
Empire Strikes Back, The (1980)	4.700000	120
Star Wars (1977)	4.617647	170
Return of the Jedi (1983)	4.500000	150
L.A. Confidential (1997)	4.388889	72
Titanic (1997)	4.368421	57
Terminator 2: Judgment Day (1991)	4.333333	54
Braveheart (1995)	4.294118	51
Fargo (1996)	4.291667	72
Alien (1979)	4.285714	56
Aliens (1986)	4.125000	64

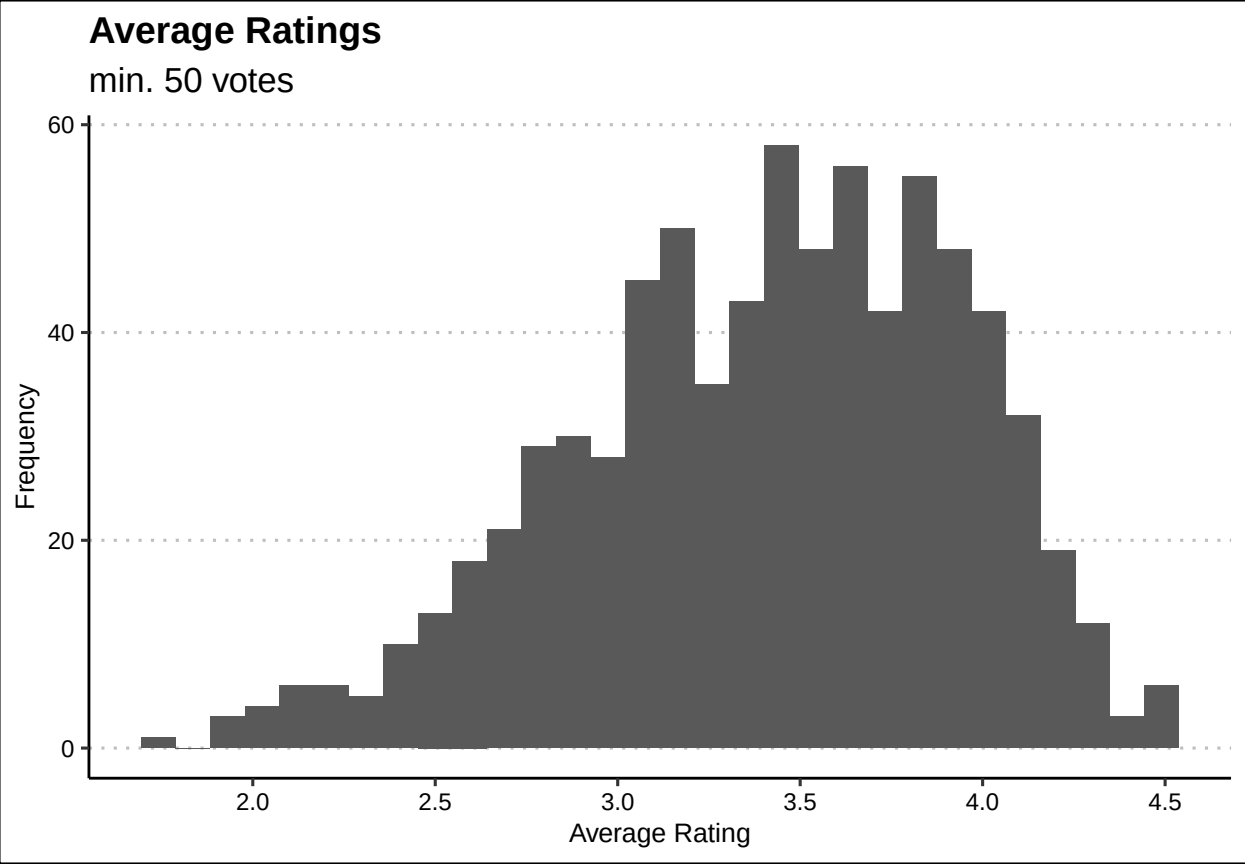
Table 8: Top 6 Movies amoung Women under 20

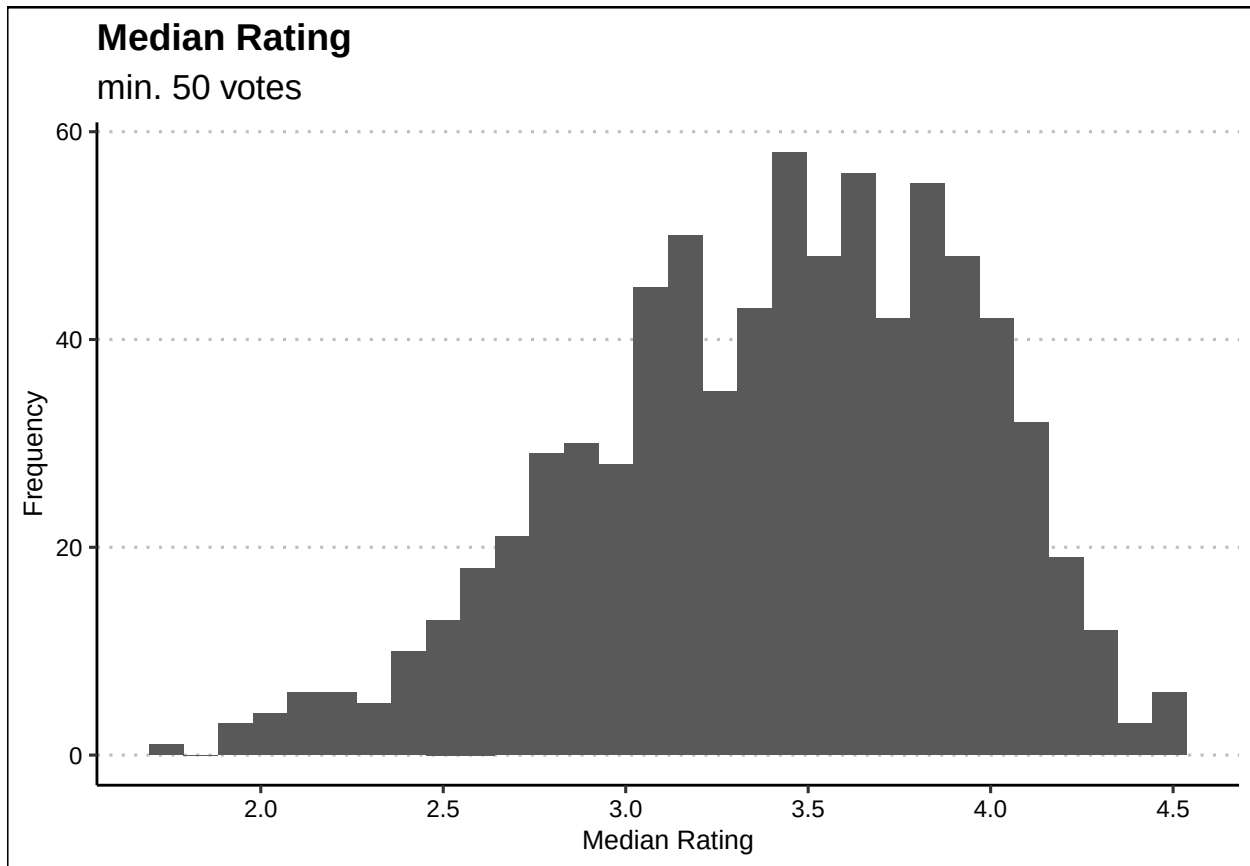
movie_title	avg_rating	rating_count
Titanic (1997)	4.888889	54
Star Wars (1977)	4.538462	65
Empire Strikes Back, The (1980)	4.444444	54
Return of the Jedi (1983)	4.428571	70
Independence Day (ID4) (1996)	4.117647	51
Scream (1996)	3.566667	60

Problem 2: Expand our investigation to histogram

Along with tables of list titles, lets expand our investigation further to ratings distributions. With this analysis, we'll be using histograms to plot the frequency of each vote, based on different audience criteria.







Problem 3: Correlation: Men versus women

Now let's dig into these ratings a bit further, breaking down by gender and age. First, let's look at the highest average movies rated by men. We've noticed that none of the average ratings in the Men's favorite movie list attains an average greater than 4.5 rating, so we'll just look at the top 10 over all, their average rating, and the number of votes the title has in the data.

Table 9: Top 10 Movies among Men, min 50 votes

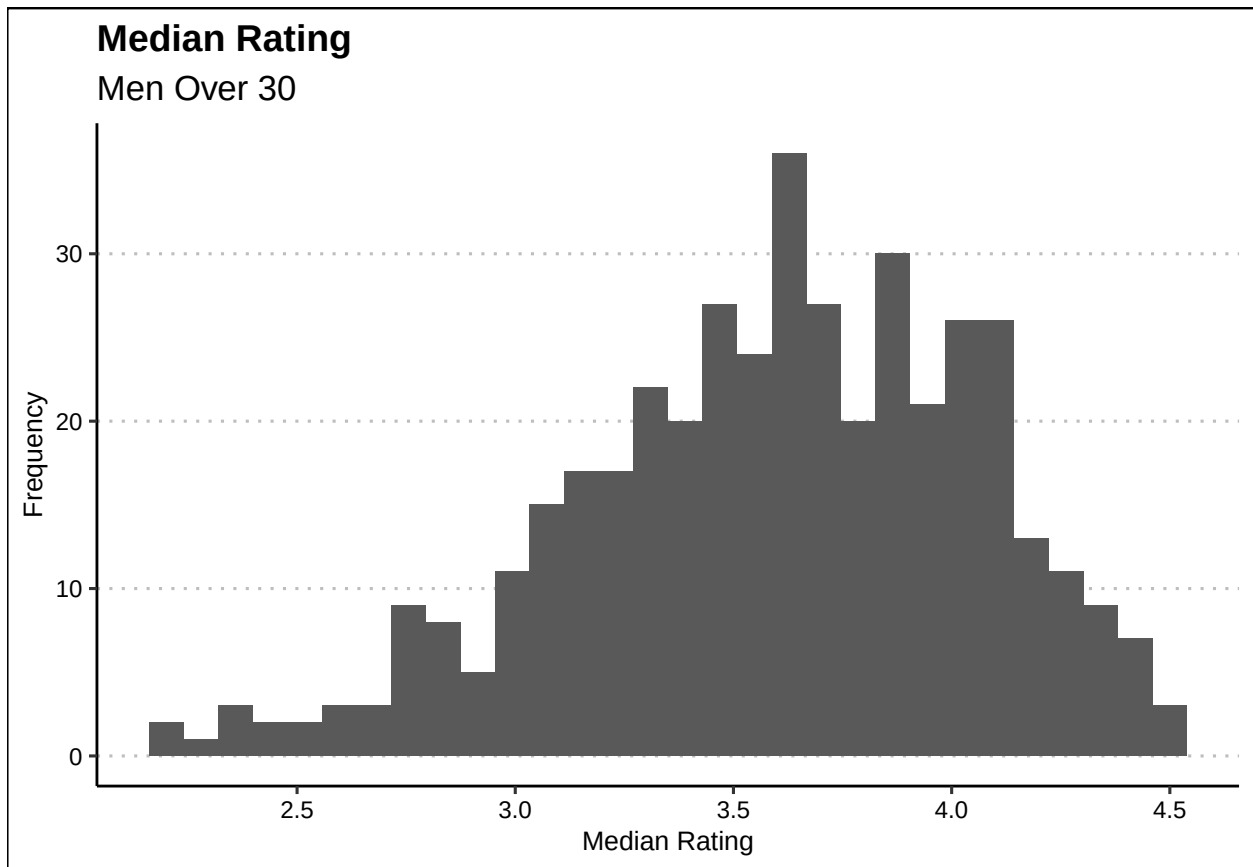
movie_title	avg_rating	rating_count
Casablanca (1942)	4.473404	564
Wrong Trousers, The (1993)	4.472527	182
Close Shave, A (1995)	4.462366	279
Third Man, The (1949)	4.433333	120
Wallace & Gromit: The Best of Aardman Animation (1996)	4.423077	52
Paths of Glory (1957)	4.419355	62
Shawshank Redemption, The (1994)	4.410959	219
Schindler's List (1993)	4.406393	438
Rear Window (1954)	4.400000	310
Usual Suspects, The (1995)	4.399061	426

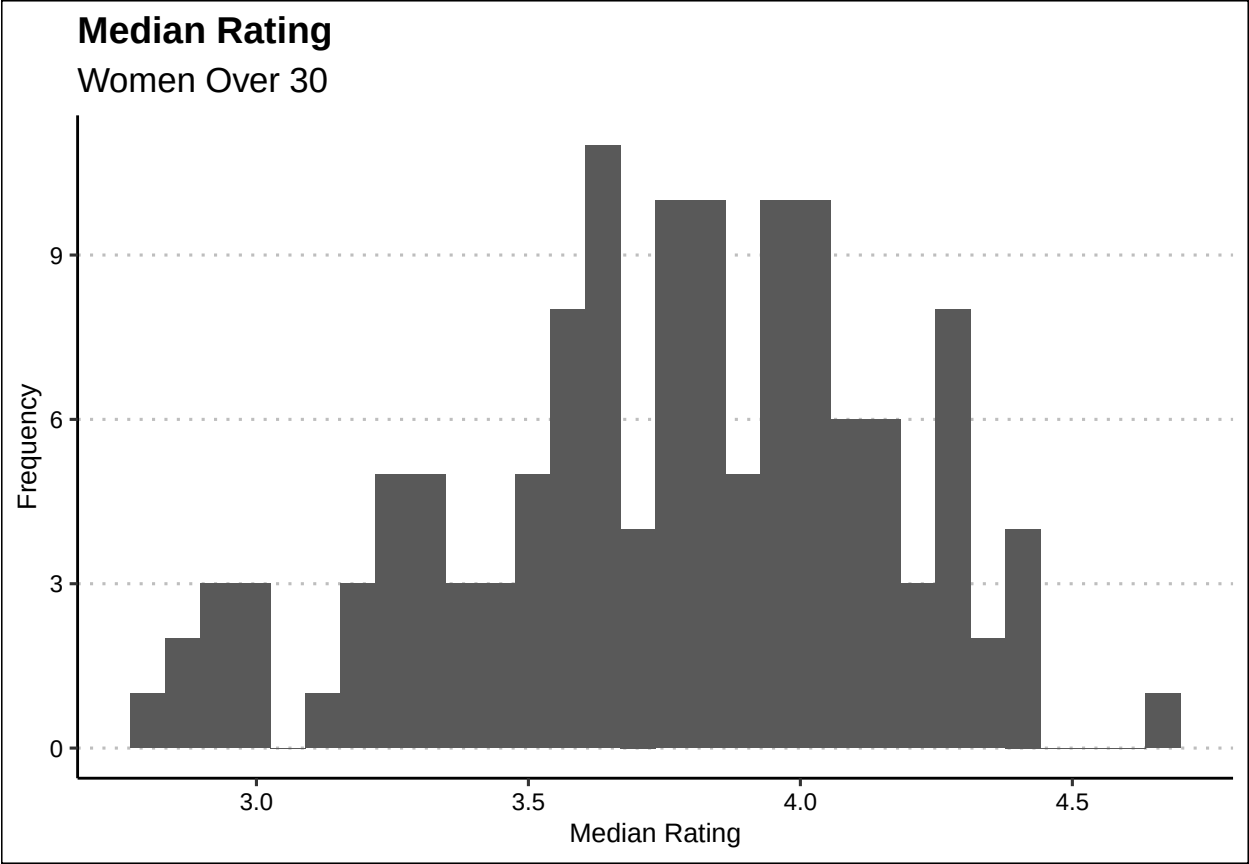
Next, we'll repeat the same analysis for the women ratings.

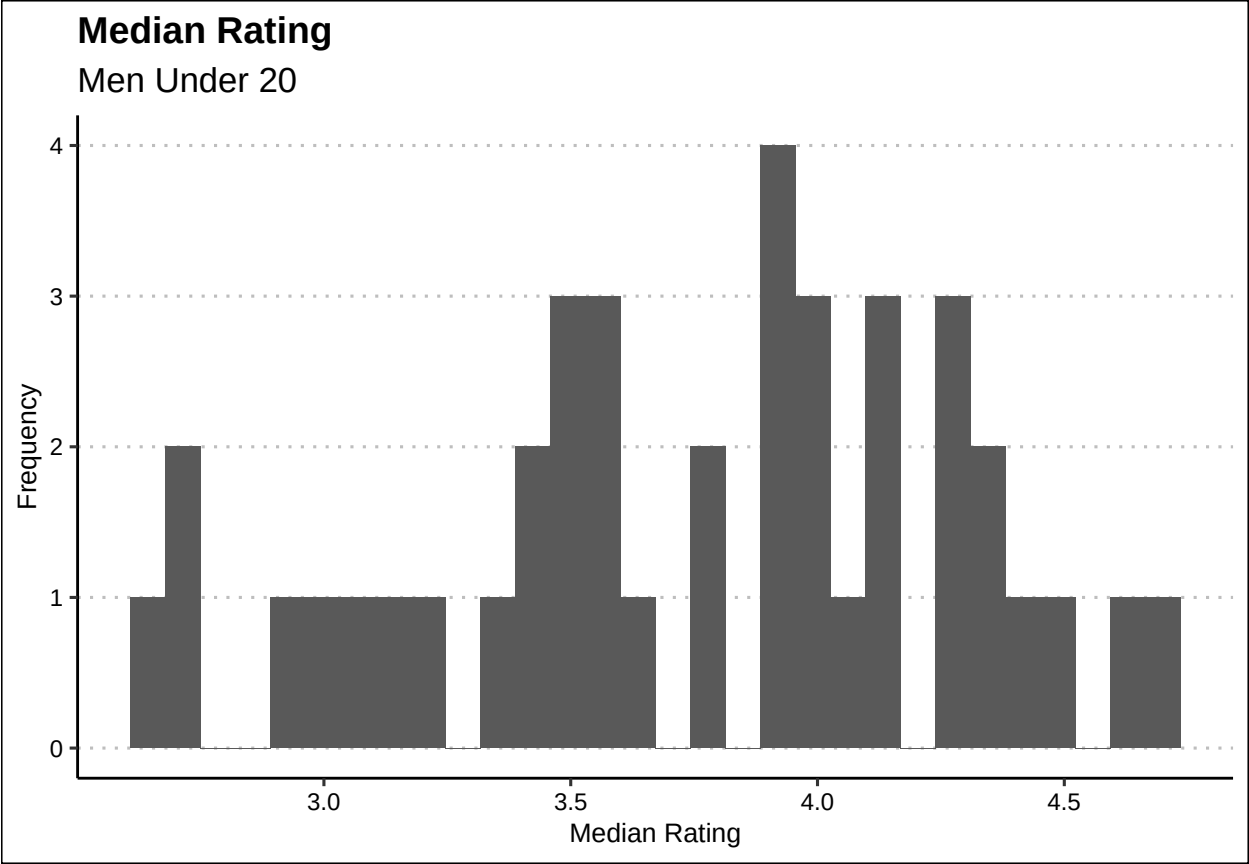
Table 10: Top 10 Movies among Women, min 50 votes

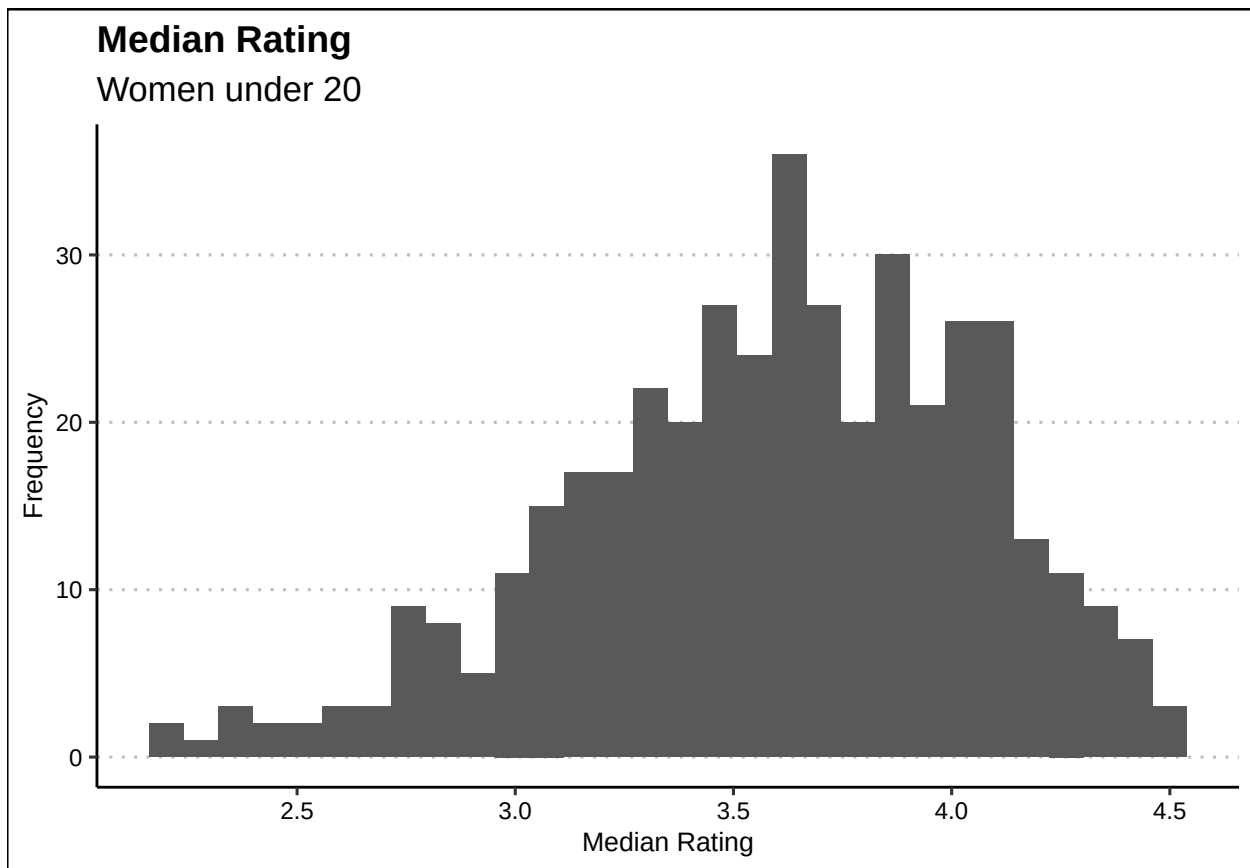
movie_title	avg_rating	rating_count
Schindler's List (1993)	4.632911	158
Close Shave, A (1995)	4.631579	57
Shawshank Redemption, The (1994)	4.562500	64
Wrong Trousers, The (1993)	4.444444	54
Sling Blade (1996)	4.411765	68
Secrets & Lies (1996)	4.407407	54
Casablanca (1942)	4.400000	165
Much Ado About Nothing (1993)	4.381818	110
Good Will Hunting (1997)	4.355932	59
Rear Window (1954)	4.351852	108

Continuing our investigation into ratings distributions, let's drill down further into possible gender preferences further. What does the distribution of ratings look like between demographics like gender and age? The following histograms seek to answer exactly that question:







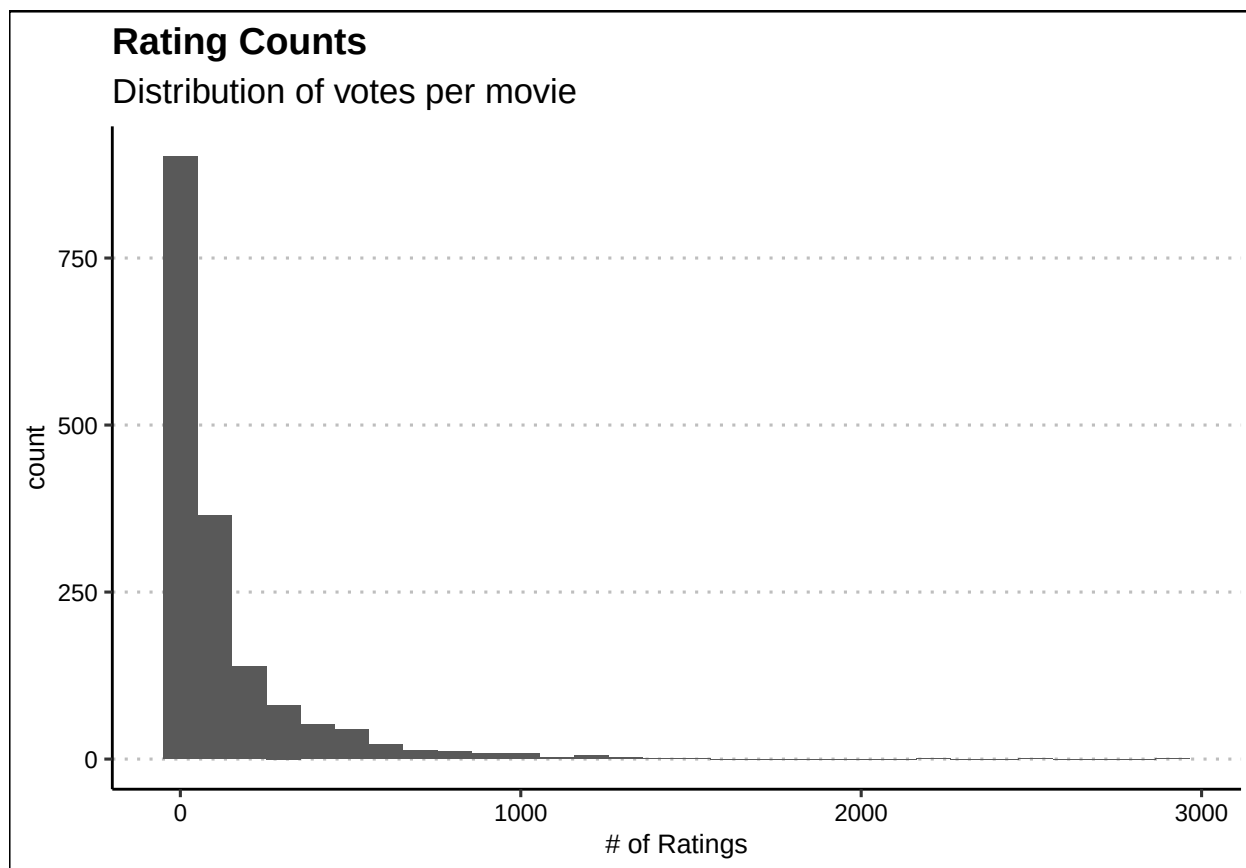


Along with the gender and age comparisons, let's take a look at the distribution of votes by title.

First, by looking at the total distribution of the ratings across all titles, we see significant clustering of voters towards a single title, which is to say only a very small number of films are receiving a high volume of votes. This tells us that very popular movies attract high numbers of voters to submit a score for that title, which is a strong indicator of popularity. We can dig into this list of popularity further by looking at the distribution of votes among the top voted titles.

Table 11: Top 10 Movies by Rating Count

movie__title	ratingCount
Star Wars (1977)	2915
Return of the Jedi (1983)	2535
Empire Strikes Back, The (1980)	2202
Fargo (1996)	1524
English Patient, The (1996)	1443
Toy Story (1995)	1356
Princess Bride, The (1987)	1296
Independence Day (ID4) (1996)	1287
Godfather, The (1972)	1239
Men in Black (1997)	1212



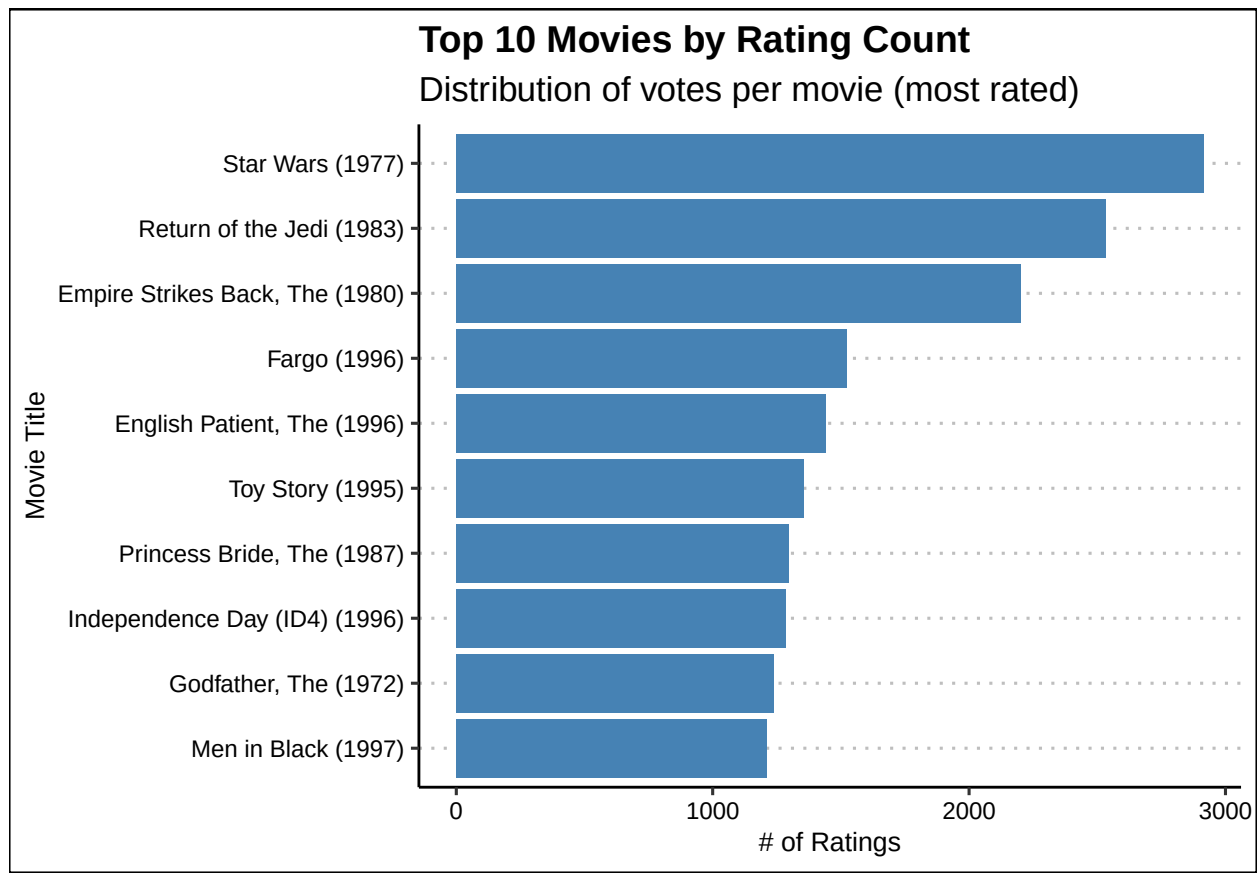
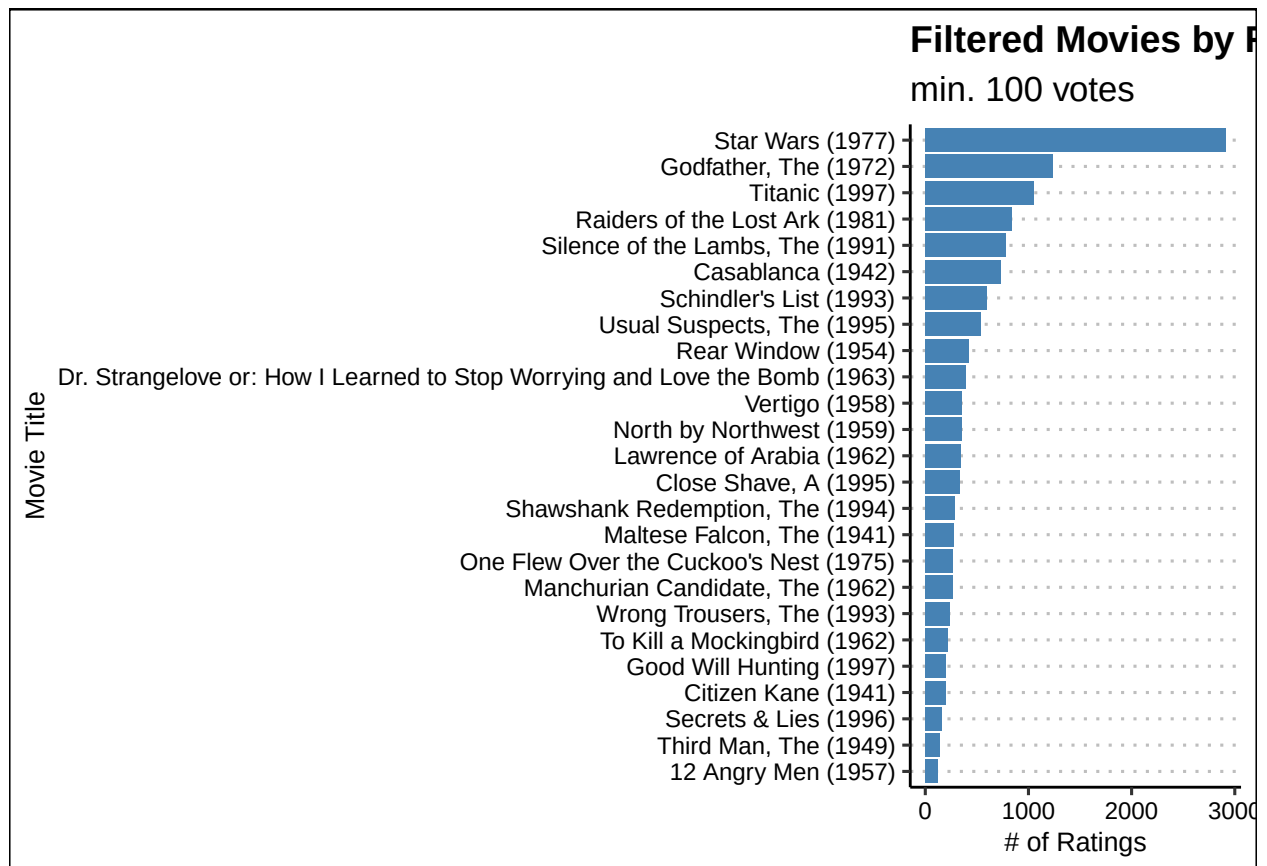


Table 12: Top 10 Movies by Rating Count, min. 100 votes

movie_title	avg_rating	rating_count
Close Shave, A (1995)	4.491071	336
Schindler's List (1993)	4.466443	596
Wrong Trousers, The (1993)	4.466102	236
Casablanca (1942)	4.456790	729
Shawshank Redemption, The (1994)	4.445230	283
Rear Window (1954)	4.387560	418
Usual Suspects, The (1995)	4.385768	534
Star Wars (1977)	4.358491	2915
12 Angry Men (1957)	4.344000	125
Third Man, The (1949)	4.333333	144



Problem 4: Open Ended Question: Business Intelligence

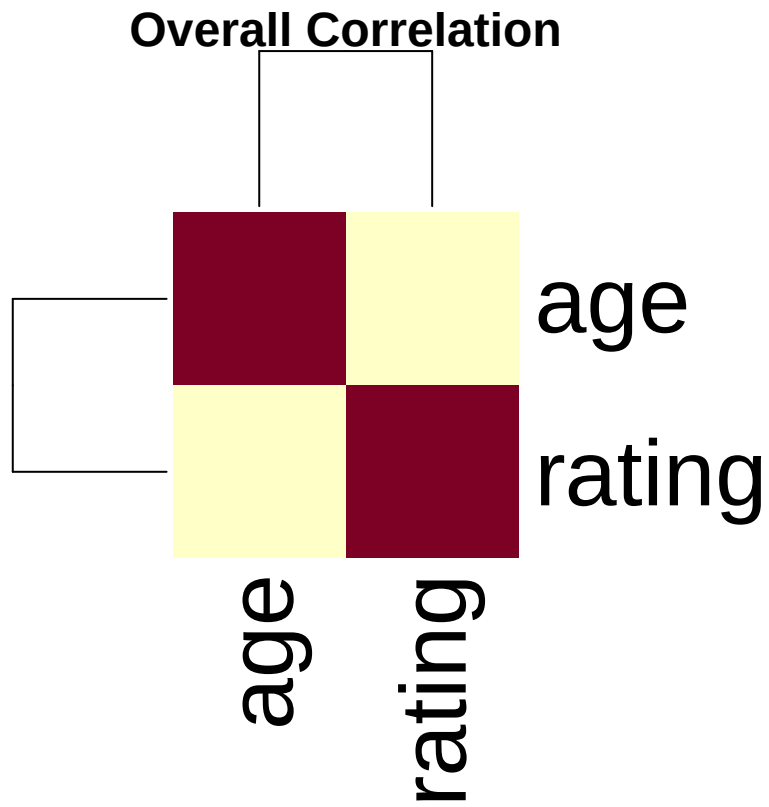
Along with histograms, let's look at some correlation heatmaps of a possible relationship between movie rating and ages:

```
##          age      rating
## age      1.00000000  0.04742223
## rating  0.04742223  1.00000000
```

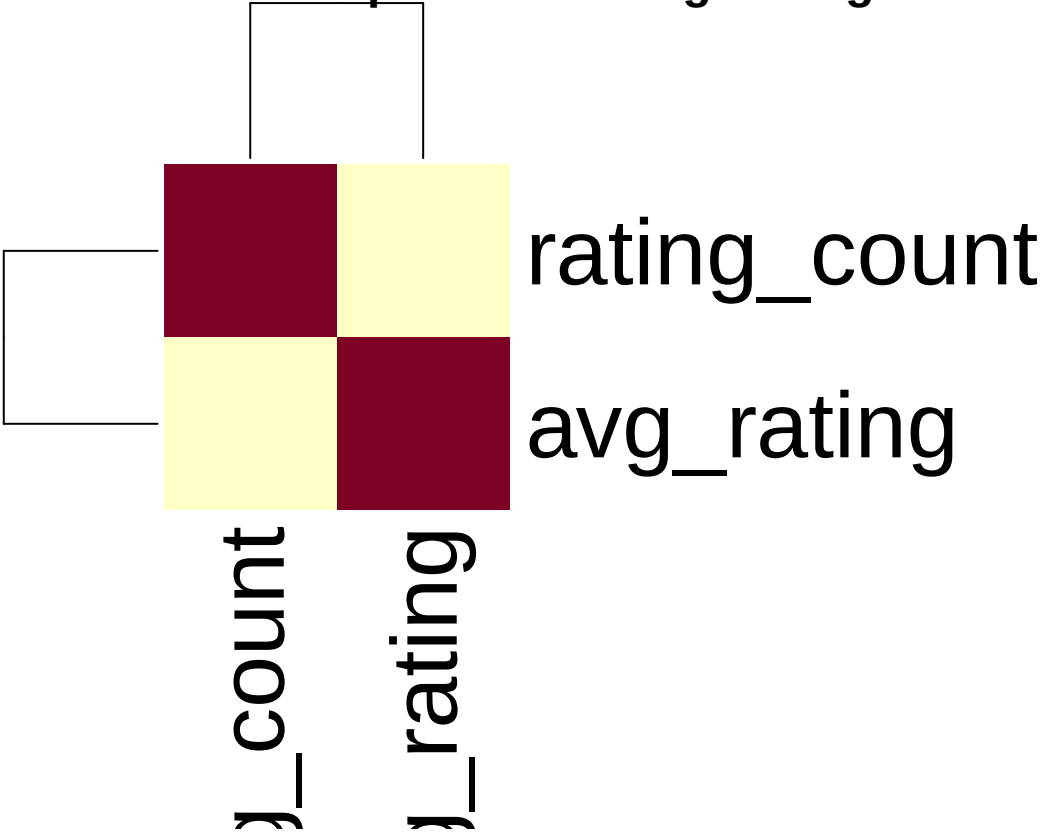
```
##          rating_count avg_rating
## rating_count  1.00000000  0.04789724
## avg_rating    0.04789724  1.00000000
```

```
##          age      rating
## age      1.00000000  0.05200609
## rating  0.05200609  1.00000000
```

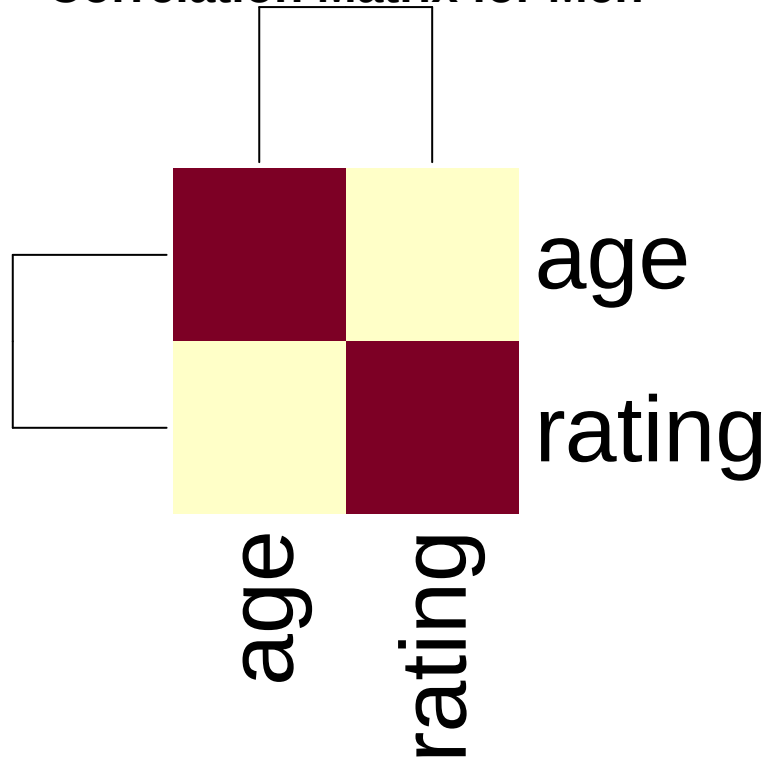
```
##          age      rating
## age      1.00000000  0.0337037
## rating  0.0337037  1.0000000
```

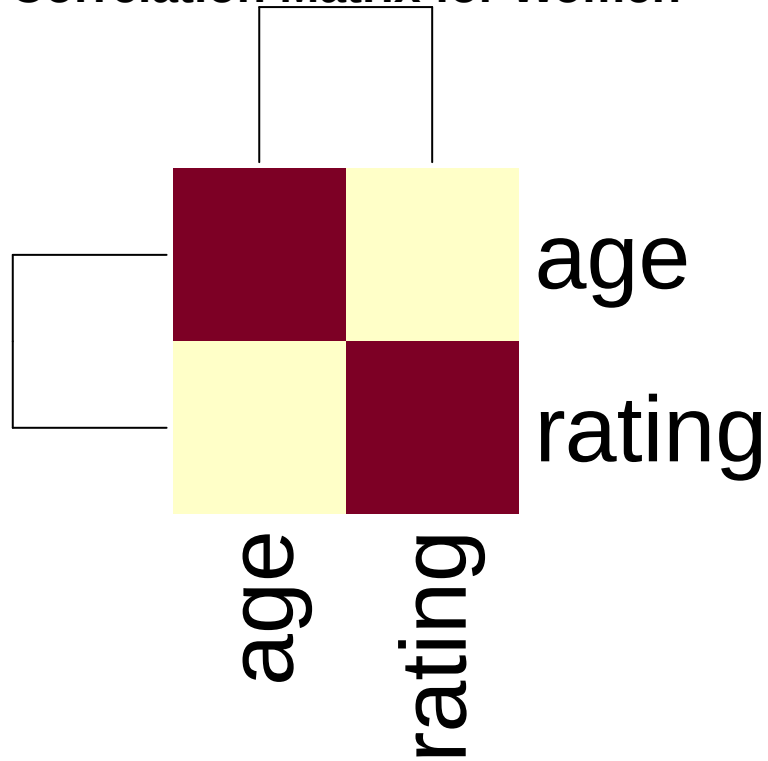
Correlation between Occupation and Avg Rating



Correlation Matrix for Men

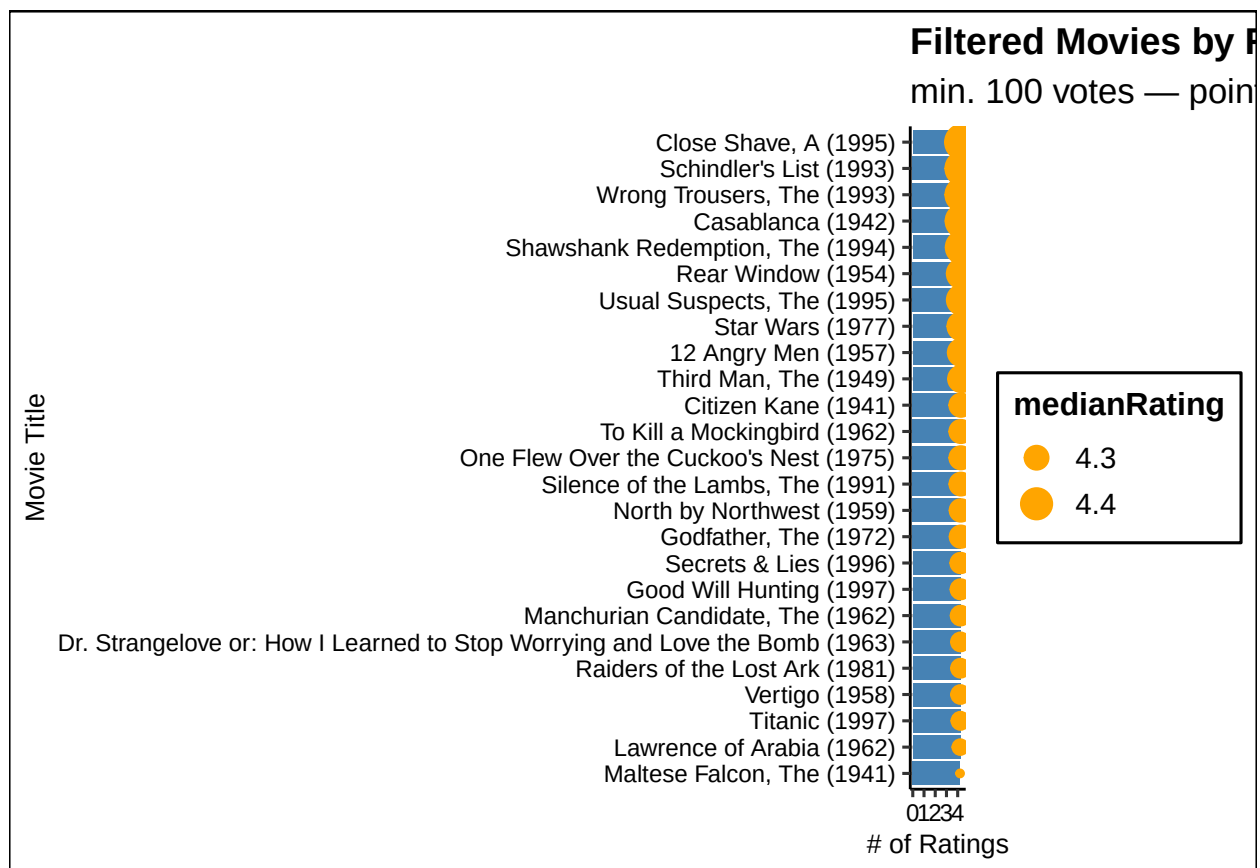


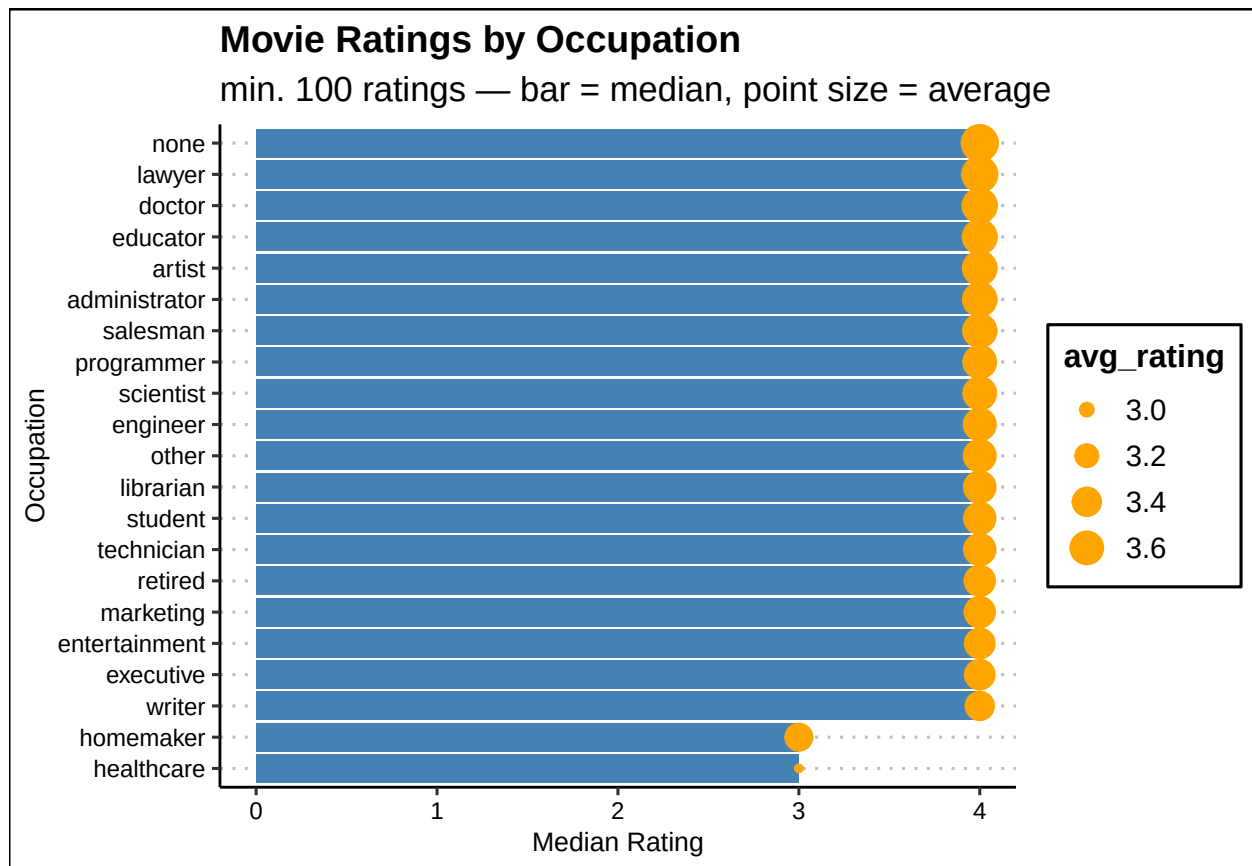
Correlation Matrix for Women



Closing Thoughts

Lastly, we'll explore some more open-ended questions with some comparisons of the top performing movies and their relative median scores.





Further business questions to investigate within the data include; further study into how genre and age interact, and what makes students either prefer movies, or prefer to review movies, relative to other occupations.